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Exercise list: L05 — Introduction to causal inference (Classes 5+6): textbook exercises

Solutions

Instructions. These are the *Practice* items of Classes 5+6 in `Map/class-plan.md`. Every question opens with a source line. Book statements (Wooldridge, Greenberg) are reproduced verbatim from the book PDF. *Mostly Harmless Econometrics* has no end-of-chapter exercises: for each of its worked examples the table is reproduced verbatim and the question is original. Computer exercises are solved in Python (`statsmodels`, `linearmodels`); `Listas/L05-book-exercises-check.py` reproduces and asserts every number in the solutions. Show your steps.

Block 1 — MHE worked examples *(MHE Ch. 2–6 • Questions 1–6)*

Question 1. The STAR experiment: balance, selection bias and covariates (MHE Tables 2.2.1–2.2.2). *Source:* Angrist and Pischke (2008), *Mostly Harmless Econometrics: An Empiricist's Companion*, §2.2, Tables 2.2.1 and 2.2.2, printed pp. 14–15. The source has no exercise; the statement is original. The table numbers are reproduced verbatim from the PDF (table notes shortened).

Table 2.2.1: Comparison of treatment and control characteristics in the Tennessee STAR experiment

		Students who entered STAR in kindergarten			
	Variable	Small	Regular	Regular/Aide	Joint P-value
1.	Free lunch	.47	.48	.50	.09
2.	White/Asian	.68	.67	.66	.26
3.	Age in 1985	5.44	5.43	5.42	.32
4.	Attrition rate	.49	.52	.53	.02
5.	Class size in kindergarten	15.10	22.40	22.80	.00
6.	Percentile score in kindergarten	54.70	48.90	50.00	.00

Notes (excerpt): "Adapted from Krueger (1999), Table 1. The table shows means of variables by treatment status. The P-value in the last column is for the F-test of equality of variable means across all three groups."

Table 2.2.2: Experimental estimates of the effect of class-size assignment on test scores

Explanatory variable	(1)	(2)	(3)	(4)
Small class	4.82	5.37	5.36	5.37
	(2.19)	(1.26)	(1.21)	(1.19)
Regular/aide class	.12	.29	.53	.31
	(2.23)	(1.13)	(1.09)	(1.07)
White/Asian (1 = yes)	–	–	8.35	8.44
			(1.35)	(1.36)
Girl (1 = yes)	–	–	4.48	4.39
			(.63)	(.63)
Free lunch (1 = yes)	–	–	-13.15	-13.07
			(.77)	(.77)
White teacher	–	–	–	-.57
				(2.10)
Teacher experience	–	–	–	.26
				(.10)
Master's degree	–	–	–	-0.51
				(1.06)
School fixed effects	No	Yes	Yes	Yes
R^2	.01	.25	.31	.31

Note (excerpt): "The dependent variable is the Stanford Achievement Test percentile score. Robust standard errors that allow for correlated residuals within classes are shown in parentheses. The sample size is 5681."

(a) What null hypothesis does the "Joint P-value" column test? Using rows 1–3 of Table 2.2.1, did randomization balance the groups at the 5% level? Why should rows 4–6 not be read as balance checks?

(b) Write the observed difference in mean scores between small and regular classes as the effect on the treated plus selection bias, and explain why random assignment makes the coefficient on *Small class* an estimate of the average causal effect.

(c) Using column (1) of Table 2.2.2, compute the t statistic and a 95% confidence interval for the small-class effect; repeat with column (4). Is the regular/aide effect in column (1) significant?

(d) From column (2) to column (3) the small-class coefficient barely moves while its standard error falls from 1.26 to 1.21. Explain both facts, and show that the fall in the standard error is about what the rise in R^2 from .25 to .31 predicts.

(e) The text says the effect is about $.2\sigma$, where σ is the standard deviation of the kindergarten percentile score. What σ does column (4) imply? Is it plausible for a percentile scale?

Notation.

- Y_i: percentile score of pupil i (0–100); $D_i \in \{0,1\}$: assigned to a small class; Y_{1i}, Y_{0i}: potential scores with and without a small class.
- $\hat{\rho}$: estimated coefficient on *Small class*; $\widehat{se}$: its standard error; $t = \hat{\rho}/\widehat{se}$; X_i: covariates (race, sex, free lunch, teacher traits).
- R^2_k: R^2 of column k; σ: standard deviation of the percentile score ($\sigma > 0$).

Step — (a) balance. For each variable the test is $H_0 : \mu_{\text{small}} = \mu_{\text{regular}} = \mu_{\text{aide}}$ (equal group means). Rows 1–3 have $p = .09, .26, .32$, all above .05, so equality is not rejected: the pre-assignment traits are balanced, as random assignment predicts. Rows 4–6 are measured *after* assignment, so they are outcomes of the treatment: row 5 (15.10 against 22.40) shows the experiment created the class-size gap (the "first stage"), row 6 is the effect itself, and row 4 ($p = .02$) flags differential attrition, a threat for later grades but not for kindergarten scores.

Step — (b) selection bias. Write $Y_i = Y_{0i} + (Y_{1i} - Y_{0i})D_i$, so the observed difference in means is

$$E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] = E[Y_{1i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 0]. \tag{Q1.1}$$

Adding and subtracting $E[Y_{0i} \mid D_i = 1]$ in (Q1.1) gives

$$E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] = \underbrace{E[Y_{1i} - Y_{0i} \mid D_i = 1]}_{\text{effect on the treated}} + \underbrace{E[Y_{0i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 0]}_{\text{selection bias}}. \tag{Q1.2}$$

Random assignment makes D_i independent of (Y_{0i}, Y_{1i}):

$$E[Y_{0i} \mid D_i = 1] = E[Y_{0i} \mid D_i = 0], \qquad E[Y_{1i} - Y_{0i} \mid D_i = 1] = E[Y_{1i} - Y_{0i}]. \tag{Q1.3}$$

Substituting (Q1.3) into (Q1.2), the selection-bias term is (0) and the first term is the average treatment effect. A regression of Y_i on a treatment dummy returns exactly the difference in group means, so $\hat{\rho}$ in column (1) estimates the average causal effect of a small class (relative to a regular class with a part-time aide).

Step — (c) inference. With $t = \hat{\rho}/\widehat{se}$ and the interval $\hat{\rho} \pm 1.96\,\widehat{se}$, column (1) gives

$$t = \frac{(4.82)}{(2.19)} = 2.2009, \tag{Q1.4}$$

$$\hat{\rho} \pm 1.96\,\widehat{se} = (4.82) \pm (1.96)(2.19) = 4.82 \pm 4.2924 = [0.5276,\ 9.1124]. \tag{Q1.5}$$

Column (4) gives

$$t = \frac{(5.37)}{(1.19)} = 4.5126, \tag{Q1.6}$$

$$\hat{\rho} \pm 1.96\,\widehat{se} = (5.37) \pm (1.96)(1.19) = 5.37 \pm 2.3324 = [3.0376,\ 7.7024]. \tag{Q1.7}$$

Both exceed 1.96, so the small-class effect is significant. The regular/aide coefficient has $t = (.12)/(2.23) = 0.0538$: no evidence that an aide matters.

Step — (d) covariates. Let X_i be the added covariates and π the slope of X_i on D_i. By the omitted-variables formula (MHE §3.2.2, pp. 44–47),

$$\rho^{\text{short}} = \rho^{\text{long}} + \gamma'\pi, \tag{Q1.8}$$

where γ is the effect of X_i on Y_i. Randomization makes $\pi \approx 0$, so by (Q1.8) the coefficient barely changes (5.37 against 5.36). The covariates do explain scores, which lowers the residual variance. With homoskedastic errors,

$$\widehat{se}(\hat{\rho}) \approx \sqrt{\frac{(1-R^2)\,V(Y_i)}{n\,V(\tilde{D}_i)}}, \tag{Q1.9}$$

where $\tilde{D}_i$ is the residual of D_i on the other regressors; since D_i is unrelated to X_i, $V(\tilde{D}_i)$ is unchanged. Dividing (Q1.9) for column (3) by (Q1.9) for column (2):

$$\frac{\widehat{se}_3}{\widehat{se}_2} \approx \sqrt{\frac{1-R_3^2}{1-R_2^2}} = \sqrt{\frac{1-(.31)}{1-(.25)}} = \sqrt{\frac{0.69}{0.75}} = 0.95917, \tag{Q1.10}$$

$$\widehat{se}_3 \approx (1.26)(0.95917) = 1.20855, \tag{Q1.11}$$

which rounds to the reported 1.21. (The table's clustered standard errors make (Q1.9) an approximation.)

Step — (e) effect size. Setting $.2\sigma$ equal to the column (4) estimate,

$$\sigma = \frac{(5.37)}{(0.2)} = 26.85. \tag{Q1.12}$$

A percentile rank spread evenly over $[0, 100]$ has standard deviation $100/\sqrt{12} = 28.8675$, so 26.85 is plausible. The raw gap in Table 2.2.1, row 6, is $(54.70) - (48.90) = 5.80$, close to the regression estimates.

Intuition. Randomization removes selection bias, so the simple comparison is already causal; covariates buy precision, not identification. If adding controls had moved $\hat{\rho}$ a lot, that would have been a sign the randomization failed.

Ref: MHE §2.1–2.3 (pp. 10–18), selection bias and regression analysis of experiments; §3.2.2 (pp. 44–47), OVB formula.

Question 2. Returns to schooling and omitted-variables bias (MHE Table 3.2.1).
Source: Angrist and Pischke (2008), *Mostly Harmless Econometrics*, §3.2.2, Table 3.2.1, printed p. 62. The source has no exercise; the statement is original. Table reproduced verbatim from the PDF.

Table 3.2.1: Estimates of the returns to education for men in the NLSY

	(1)	(2)	(3)	(4)	(5)
Controls:	None	Age dummies	Col. (2) and additional controls*	Col. (3) and AFQT score	Col. (4), with occupation dummies
	0.132	0.131	0.114	0.087	0.066
	(0.007)	(0.007)	(0.007)	(0.009)	(0.010)

Notes: "Data are from the National Longitudinal Survey of Youth (1979 cohort, 2002 survey). The table reports the coefficient on years of schooling in a regression of log wages on years of schooling and the indicated controls. Standard errors are shown in parentheses. The sample is restricted to men and weighted by NLSY sampling weights. The sample size is 2434. *Additional controls are mother's and father's years of schooling and dummy variables for race and Census region."

(a) Write the long regression of log wages on schooling, the column (3) controls and AFQT, and the auxiliary regression of AFQT on schooling and the same controls. Derive the omitted-variables-bias (OVB) formula that links the schooling coefficients in columns (3) and (4).

(b) Compute the change in the schooling coefficient attributable to AFQT, and to the family-background controls (column (2) to column (3)). What do the signs say about the correlations involved?

(c) Suppose the auxiliary regression of AFQT (in standard-deviation units) on schooling, with the column (3) controls, had slope 0.15 (a hypothetical value, not in the book). What effect of one standard deviation of AFQT on log wages does the OVB formula then imply?

(d) Column (5) is half of column (1). Why is column (5) *not* the most credible estimate of the return to schooling?

(e) Give a 95% confidence interval for the return to schooling in column (4).

Notation.

- Y_i: log wage; S_i: years of schooling; A_i: AFQT score (standard-deviation units); X_i: the column (3) controls.
- ρ: schooling coefficient in the long regression (with A_i); ρ^s: in the short one (without A_i); γ: AFQT coefficient; δ_{AS}: slope of A_i on S_i given X_i.

Step — (a) OVB formula. The long, auxiliary and short regressions are

$$Y_i = \alpha + \rho S_i + X_i'\beta + \gamma A_i + e_i, \tag{Q2.1}$$

$$A_i = \delta_0 + \delta_{AS} S_i + X_i'\pi + v_i, \tag{Q2.2}$$

$$Y_i = \alpha^s + \rho^s S_i + X_i'\beta^s + e_i^s. \tag{Q2.3}$$

Substituting (Q2.2) into (Q2.1):

$$Y_i = \alpha + \rho S_i + X_i'\beta + \gamma\left(\delta_0 + \delta_{AS} S_i + X_i'\pi + v_i\right) + e_i. \tag{Q2.4}$$

Collecting terms in (Q2.4):

$$Y_i = (\alpha + \gamma\delta_0) + (\rho + \gamma\delta_{AS}) S_i + X_i'(\beta + \gamma\pi) + (\gamma v_i + e_i). \tag{Q2.5}$$

The error $\gamma v_i + e_i$ is uncorrelated with S_i and X_i (v_i is an OLS residual, e_i is the long-regression residual), so (Q2.5) is the short regression (Q2.3) and

$$\rho^s = \rho + \gamma\,\delta_{AS}. \tag{Q2.6}$$

Step — (b) the numbers. Column (3) is the short regression and column (4) the long one. From (Q2.6):

$$\gamma\,\delta_{AS} = \rho^s - \rho = (0.114) - (0.087) = 0.027. \tag{Q2.7}$$

The bias is positive, so γ and δ_{AS} have the same sign: AFQT is positively related to both wages and schooling. The family-background block lowers the coefficient by $(0.131) - (0.114) = 0.017$, again positive: parents' schooling raises both the son's schooling and his wage.

Step — (c) implied AFQT effect. Solving (Q2.7) for γ with $\delta_{AS} = 0.15$:

$$\gamma = \frac{0.027}{\delta_{AS}} = \frac{0.027}{(0.15)} = 0.18, \tag{Q2.8}$$

i.e. one standard deviation of AFQT would raise wages by about 18 log points, holding schooling fixed.

Step — (d) bad control. Occupation is itself an outcome of schooling. Conditioning on it compares men within occupations, which (i) removes a channel through which schooling raises wages and (ii) creates selection bias, because the men with low schooling who reach a given occupation differ in unobservables from those with high schooling in the same occupation. Column (5), $0.066/0.132 = 0.50$ of column (1), mixes these two effects and has no causal interpretation. Column (4) is preferred, keeping in mind that AFQT may itself be partly affected by schooling (a "proxy control").

Step — (e) interval. Column (4): $(0.087) \pm (1.96)(0.009) = 0.087 \pm 0.01764 = [0.06936,\ 0.10464]$.

Intuition. Each block of controls lowers the coefficient because each omitted factor is positively correlated with both schooling and wages. The OVB formula tells you the direction of the bias before you run anything; the bad-control argument tells you where to stop adding controls.

Ref: MHE §3.2.2 (pp. 44–47), OVB formula; §3.2.3 (pp. 47–50), bad control; W §3.3, omitted-variable bias.

Question 3. Wald and 2SLS estimates: quarter of birth and the draft lottery (MHE Tables 4.1.1–4.1.3).

Source: Angrist and Pischke (2008), *Mostly Harmless Econometrics*, §4.1, Tables 4.1.1, 4.1.2 and 4.1.3, printed pp. 92, 96 and 97. The source has no exercise; the statement is original. Table numbers reproduced verbatim from the PDF (notes shortened).

Table 4.1.1: 2SLS estimates of the economic returns to schooling

	OLS		2SLS					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Years of education	0.075	0.072	0.103	0.112	0.106	0.108	0.089	0.061
	(0.0004)	(0.0004)	(0.024)	(0.021)	(0.026)	(0.019)	(0.016)	(0.031)
Covariates:								
Age (in quarters)								✓
Age (in quarters) squared								✓
9 year of birth dummies		✓			✓	✓	✓	✓
50 state of birth dummies		✓			✓	✓	✓	✓
Instruments:			dummy for QOB=1	dummy for QOB=1 or QOB=2	dummy for QOB=1	full set of QOB dummies	full set of QOB dummies int. with year of birth dummies	full set of QOB dummies int. with year of birth dummies

Notes (excerpt): "The table reports OLS and 2SLS estimates of the returns to schooling using the the Angrist and Krueger (1991) 1980 Census sample. [. . .] The sample size is 329,509. Robust standard errors are reported in parentheses."

Table 4.1.2: Wald estimates of the returns to schooling using quarter of birth instruments

	(1) Born in the 1st or 2nd quarter of year	(2) Born in the 3rd or 4th quarter of year	(3) Difference (std. error) (1)-(2)
ln (weekly wage)	5.8916	5.9051	-0.01349
			(0.00337)
Years of education	12.6881	12.8394	-0.1514
			(0.0162)
Wald estimate of return to education			0.0891
			(0.0210)
OLS estimate of return to education			0.0703
			(0.0005)

Notes (excerpt): "Adapted from a re-analysis of Angrist and Krueger (1991) by Angrist and Imbens (1995). [. . .] The sample size is 329,509."

Table 4.1.3: Wald estimates of the effects of military service on the earnings of white men born in 1950

	Earnings		Veteran Status		Wald Estimate
Earnings year	Mean (1)	Eligibility Effect (2)	Mean (3)	Eligibility Effect (4)	of Veteran Effect (5)
1981	16,461	-435.8	0.267	0.159	-2,741
		(210.5)		(0.040)	(1,324)
1971	3,338	-325.9			-2050
		(46.6)			(293)
1969	2,299	-2.0			
		(34.5)			

Notes (excerpt): "Adapted from Angrist (1990), Tables 2 and 3. Standard errors are shown in parentheses. [...] Figures are in nominal dollars. [...] There are about 13,500 individuals in the sample."

(a) Derive the Wald estimator for a binary instrument and state the assumptions under which it identifies a causal effect.

(b) Table 4.1.2: compute the Wald estimate from column (3) and check it against 0.0891. Approximate its standard error treating the first stage as known, and compare with 0.0210.

(c) Table 4.1.3: compute the Wald estimates for 1981 and 1971 from columns (2) and (4), their approximate standard errors, and each as a share of mean earnings. What is the 1969 row for?

(d) Table 4.1.1: is the 2SLS estimate in column (3) significantly different from the OLS estimate in column (1)? Use a Hausman-type statistic with $V(\hat{\rho}_{IV} - \hat{\rho}_{OLS}) = V(\hat{\rho}_{IV}) - V(\hat{\rho}_{OLS})$. Why is the standard error in column (8) larger than in columns (6)–(7), even though (8) uses as many instruments?

Notation.

- Y_i: outcome (log weekly wage, or annual earnings in dollars); D_i: treatment (years of schooling, or veteran status $\in \{0,1\}$); $Z_i \in \{0,1\}$: instrument (born in quarters 1–2, or draft-eligible).
- ρ: causal effect of D_i; η_i: structural error; $\pi = E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0]$: first stage; $\phi = E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0]$: reduced form.

Step — (a) Wald estimator. Take the constant-effect model and the instrument assumption

$$Y_i = \alpha + \rho D_i + \eta_i, \qquad E[\eta_i \mid Z_i] = E[\eta_i]. \tag{Q3.1}$$

Conditioning (Q3.1) on $Z_i = z$:

$$E[Y_i \mid Z_i = z] = \alpha + \rho\, E[D_i \mid Z_i = z] + E[\eta_i]. \tag{Q3.2}$$

Subtracting (Q3.2) at $z = 0$ from (Q3.2) at $z = 1$:

$$\phi = \rho\, \pi. \tag{Q3.3}$$

Dividing (Q3.3) by $\pi \neq 0$:

$$\rho = \frac{\phi}{\pi} = \frac{E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0]}{E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0]}. \tag{Q3.4}$$

Assumptions: Z_i as good as randomly assigned (independence), Z_i affects Y_i only through D_i (exclusion), and $\pi \neq 0$ (first stage). With heterogeneous effects and monotonicity (no defiers), (Q3.4) is the local average treatment effect on compliers (MHE §4.4).

Step — (b) quarter of birth. Substituting column (3) of Table 4.1.2 into (Q3.4):

$$\hat{\rho} = \frac{(-0.01349)}{(-0.1514)} = 0.089102, \tag{Q3.5}$$

which rounds to the reported 0.0891. (The rounded means give $5.8916 - 5.9051 = -0.0135$ and $12.6881 - 12.8394 = -0.1513$.) Treating π as known, $\widehat{se}(\hat{\rho}) \approx \widehat{se}(\hat{\phi})/|\hat{\pi}|$:

$$\widehat{se}(\hat{\rho}) \approx \frac{(0.00337)}{(0.1514)} = 0.022259, \tag{Q3.6}$$

close to 0.0210; the delta method adds the sampling noise in $\hat{\pi}$ and its covariance with $\hat{\phi}$. The Wald estimate (0.0891) exceeds OLS (0.0703).

Step — (c) draft lottery. Substituting columns (2) and (4) of Table 4.1.3 into (Q3.4):

$$\hat{\rho}_{1981} = \frac{(-435.8)}{(0.159)} = -2740.88, \tag{Q3.7}$$

$$\hat{\rho}_{1971} = \frac{(-325.9)}{(0.159)} = -2049.69, \tag{Q3.8}$$

matching −2,741 and −2050. Dividing the reduced-form standard errors by the first stage, as in (Q3.6): $(210.5)/(0.159) = 1323.90$ and $(46.6)/(0.159) = 293.08$, matching (1,324) and (293). As shares of mean earnings: $2740.88/16461 = 0.16651$ in 1981 and $2049.69/3338 = 0.61405$ in 1971, so service cut earnings by about 17% ten years later and by much more while the men were still in the army. The 1969 row is a placebo: earnings before anyone born in 1950 was drafted cannot depend on veteran status, and indeed $t = (-2.0)/(34.5) = -0.0580$. A zero there supports the exclusion restriction.

Step — (d) OLS against 2SLS. The Hausman-type statistic with the given variance:

$$H = \frac{(0.103) - (0.075)}{\sqrt{(0.024)^2 - (0.0004)^2}} = \frac{0.028}{0.023997} = 1.1668 < 1.96, \tag{Q3.9}$$

so the difference is not significant: quarter of birth is too weak a source of variation to separate the two. Column (8) adds age in quarters and its square. Within a year of birth, age in quarters is a linear function of quarter of birth, so these controls absorb much of the instrument's variation: the first stage weakens and the standard error rises (0.031 against 0.016–0.019). Columns (7)–(8) also use 30 instruments, which pulls 2SLS towards OLS when the instruments are weak (the many-weak-instruments bias discussed in MHE Ch. 4).

Intuition. The Wald estimator rescales the effect of the instrument on the outcome by its effect on the treatment. The QOB first stage is tiny (0.15 years), so small reduced-form effects translate into imprecise IV estimates; the draft first stage is larger (16 points), so the estimates are informative.

Ref: MHE §4.1 (pp. 84–102), IV and the Wald estimator; §4.4 (pp. 111–129), LATE; W §15.1 (pp. 462–471).

Question 4. Card and Krueger: difference-in-differences by hand (MHE Table 5.2.1).
Source: Angrist and Pischke (2008), *Mostly Harmless Econometrics*, §5.2, Table 5.2.1, printed p. 171. The source has no exercise; the statement is original. Table reproduced verbatim from the PDF.

Table 5.2.1: Average employment per store before and after the New Jersey minimum wage increase

	Variable	PA (i)	NJ (ii)	Difference, NJ-PA (iii)
1.	FTE employment before,	23.33	20.44	-2.89
	all available observations	(1.35)	(0.51)	(1.44)
2.	FTE employment after,	21.17	21.03	-0.14
	all available observations	(0.94)	(0.52)	(1.07)
3.	Change in mean FTE	-2.16	0.59	2.76
	employment	(1.25)	(0.54)	(1.36)

Notes (excerpt): "Adapted from Card and Krueger (1994), Table 3. The table reports average full-time equivalent (FTE) employment at restaurants in Pennsylvania and New Jersey before and after a minimum wage increase in New Jersey. [. . .] Standard errors are reported in parentheses"

(a) Compute the difference-in-differences estimate twice: as the difference of the two states' changes, and as the change in the NJ–PA gap. Reconcile with 2.76.

(b) Write the regression $Y_{ist} = \alpha + \gamma NJ_s + \lambda d_t + \delta(NJ_s \cdot d_t) + \varepsilon_{ist}$ and give the value of each coefficient implied by the table. State the identifying assumption.

(c) Compute the t statistic and a 95% interval for the DiD estimate, and show that its standard error is what independent samples in the two states imply.

(d) Is the sign of the estimate what a competitive labor-demand model predicts?

Notation.

- Y_{ist}: FTE employment of store i in state $s \in \{PA, NJ\}$ at $t \in \{\text{before}, \text{after}\}$; $\bar{Y}_{st}$: cell mean; Y_{0ist}: employment without the minimum-wage rise.
- NJ_s, d_t: dummies for New Jersey and for the after period; γ_s, λ_t: state and time effects; δ: causal effect of the minimum wage on FTE employment.

Step — (a) two ways. The DiD estimator is

$$\hat{\delta} = (\bar{Y}_{NJ,a} - \bar{Y}_{NJ,b}) - (\bar{Y}_{PA,a} - \bar{Y}_{PA,b}). \tag{Q4.1}$$

Substituting rows 1–2:

$$\hat{\delta} = \big((21.03) - (20.44)\big) - \big((21.17) - (23.33)\big) \tag{Q4.2}$$

$$= (0.59) - (-2.16) = 2.75. \tag{Q4.3}$$

Rearranging (Q4.1) as the change in the gap and substituting column (iii):

$$\hat{\delta} = (\bar{Y}_{NJ,a} - \bar{Y}_{PA,a}) - (\bar{Y}_{NJ,b} - \bar{Y}_{PA,b}) = (-0.14) - (-2.89) = 2.75. \tag{Q4.4}$$

The table's 2.76 comes from unrounded cell means.

Step — (b) regression. Common trends means

$$E[Y_{0ist} \mid s, t] = \gamma_s + \lambda_t. \tag{Q4.5}$$

With $Y_{ist} = Y_{0ist} + \delta\, NJ_s d_t$, the NJ change is $(\lambda_a - \lambda_b) + \delta$ and the PA change is $\lambda_a - \lambda_b$; their difference is δ, which is what (Q4.1) estimates. In the regression the four cells are α, $\alpha + \gamma$, $\alpha + \lambda$, $\alpha + \gamma + \lambda + \delta$, so

$$\hat{\alpha} = 23.33, \quad \hat{\gamma} = (20.44) - (23.33) = -2.89, \quad \hat{\lambda} = (21.17) - (23.33) = -2.16, \quad \hat{\delta} = 2.75. \tag{Q4.6}$$

The identifying assumption is (Q4.5): without the rise, NJ employment would have moved like PA employment.

Step — (c) inference. If the two states' samples are independent, the variance of a difference is the sum of the variances:

$$\widehat{se}(\hat{\delta}) = \sqrt{(1.25)^2 + (0.54)^2} = \sqrt{1.5625 + 0.2916} = \sqrt{1.8541} = 1.36165, \tag{Q4.7}$$

the reported 1.36. Then $t = (2.76)/(1.36) = 2.0294$ and $2.76 \pm (1.96)(1.36) = 2.76 \pm 2.6656 = [0.0944,\ 5.4256]$: significant at 5%, but the interval is wide.

Step — (d) interpretation. A competitive model predicts lower employment in NJ, so $\delta < 0$. The estimate is positive: employment fell in PA and rose slightly in NJ. Either the market is not competitive (monopsony), or common trends fails; MHE shows that longer payroll data from the same states weaken the original result, which is why pre-period trends should be checked.

Intuition. DiD uses the control state's change as the counterfactual change for the treated state; the state effect γ absorbs any permanent NJ–PA gap, so only differential trends can bias $\hat{\delta}$.

Ref: MHE §5.2 (pp. 169–181), differences-in-differences; W §13.2 (pp. 406–412).

Question 5. Incumbency and the sharp regression-discontinuity design (MHE Figure 6.1.2).
Source: Angrist and Pischke (2008), *Mostly Harmless Econometrics*, §6.1, Figure 6.1.2 and text, printed pp. 194–195. The source has no exercise; the statement is original. Caption and numbers from the text reproduced verbatim; the two fitted values at the cutoff are read off the figure (not printed in the book).

Figure 6.1.2 (caption, verbatim): "Probability of winning an election by past and future vote share (from Lee, 2008). (a) Candidate's probability of winning election $t+1$, by margin of victory in election t: local averages and parametric fit. (b) Candidate's accumulated number of past election victories, by margin of victory in election t: local averages and parametric fit." The text states that the winner is determined by $D_i = 1(x_i \geq .0)$, where x_i is the vote share margin of victory; that the dots are "the average win rate in non-overlapping windows of share margins that are .005 wide"; that the lines are fitted from "a Logit model for the probability of winning as a function of the cutoff indicator $D_i = 1(x_i \geq 0)$, a 4th-order polynomial in x_i, and interactions between the polynomial terms and D_i"; and that "incumbency appears to raise party re-election probabilities by about 40 percentage points." In panel (a) the fitted curve reaches about 0.14 just left of zero and about 0.58 just right of zero; in panel (b) the two fitted curves nearly meet at zero (both close to one past victory).

(a) Write the sharp RD model with a polynomial control and define the estimand as a jump in a conditional expectation. State the identifying assumption.

(b) Estimate the incumbency effect from panel (a) and compare it with "about 40 percentage points".

(c) What does panel (b) test, and what does it show?

(d) Give two reasons why an RD estimate could be spurious here, and a check for each.

Notation.

- x_i: Democratic vote-share margin in election t (running variable, cutoff 0); $D_i = 1(x_i \geq 0)$: Democrat won election t (incumbency).
- Y_i: Democrat wins election $t+1$; Y_{0i}, Y_{1i}: potential outcomes; $f(\cdot)$: smooth function of x_i (a polynomial); ρ: incumbency effect at the cutoff.

Step — (a) model. Assume $E[Y_{0i} \mid x_i] = f(x_i)$ and a constant effect ρ:

$$Y_i = f(x_i) + \rho D_i + \eta_i, \qquad D_i = 1(x_i \geq 0). \tag{Q5.1}$$

Taking limits of $E[Y_i \mid x_i]$ in (Q5.1) from each side of 0, with f continuous:

$$\lim_{x \downarrow 0} E[Y_i \mid x_i = x] = f(0) + \rho, \tag{Q5.2}$$

$$\lim_{x \uparrow 0} E[Y_i \mid x_i = x] = f(0). \tag{Q5.3}$$

Subtracting (Q5.3) from (Q5.2):

$$\rho = \lim_{x \downarrow 0} E[Y_i \mid x_i = x] - \lim_{x \uparrow 0} E[Y_i \mid x_i = x]. \tag{Q5.4}$$

Identifying assumption: $E[Y_{0i} \mid x_i]$ and $E[Y_{1i} \mid x_i]$ are continuous at 0. Because D_i is a deterministic function of x_i, there is no other confounder once $f(x_i)$ is controlled for.

Step — (b) the jump. Substituting the fitted values at the cutoff into (Q5.4):

$$\hat{\rho} \approx (0.58) - (0.14) = 0.44, \tag{Q5.5}$$

an incumbency advantage of roughly 40 percentage points, as the text says.

Step — (c) panel (b). Past victories are determined before election t, so they cannot be caused by D_i. If close elections are as good as random, their conditional mean must be continuous at 0: (Q5.4)

applied to this variable should be zero. The two fitted curves nearly meet at 0, so there is no visible jump: this balance check supports the design, as a covariate-balance table does in an experiment.

Step — (d) threats. (i) Misspecified f: a nonlinear $E[Y_{0i} \mid x_i]$ fitted with too rigid a polynomial can show up as a jump. Check: higher-order and side-specific polynomials (the interactions with D_i in the footnote), and local estimates in narrow windows around 0. (ii) Manipulation: if candidates could sort just above 0 (recounts, fraud), units on the two sides would differ. Check: the density of x_i near 0 (McCrary test) and balance of pre-determined covariates such as panel (b).

Intuition. Bare winners and bare losers are alike in everything but the election outcome, so the discontinuity at a zero margin isolates what winning itself does to the next election.

Ref: MHE §6.1 (pp. 189–196), sharp RD; class slides and site example on RDD (minimum drinking age and motor-vehicle deaths).

Question 6. Maimonides' rule and fuzzy RD as IV (MHE Table 6.2.1).

Source: Angrist and Pischke (2008), *Mostly Harmless Econometrics*, §6.2, Table 6.2.1 and text, printed pp. 199–202. The source has no exercise; the statement is original. Table and the rule reproduced verbatim from the PDF.

The text defines the predicted class size from Maimonides' rule (class size capped at 40) as

$$m_{sc} = \frac{e_s}{\text{int}\left[\frac{(e_s-1)}{40}\right] + 1},$$

"where int(x) is the integer part of a real number, x", e_s is enrollment in the grade, and estimates by 2SLS "an equation like $y_{isc} = \alpha_0 + \alpha_1 d_s + \beta_1 e_s + \beta_2 e_s^2 + \cdots + \beta_p e_s^p + \rho n_{sc} + \eta_{isc}$" (6.2.6), with n_{sc} class size, d_s the proportion of disadvantaged students, and m_{sc} as instrument.

Table 6.2.1: OLS and fuzzy RD estimates of the effects of class size on fifth grade math scores

	OLS			2SLS				
				Full sample		Discontinuity samples		
						+/- 5		+/- 3
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Mean score		67.3		67.3		67.0		67.0
(s.d.)		(9.6)		(9.6)		(10.2)		(10.6)
Regressors								
Class size	.322	.076	.019	-.230	-.261	-.185	-.443	-.270
	(.039)	(.036)	(.044)	(.092)	(.113)	(.151)	(.236)	(.281)
Percent disadvantaged		-.340	-.332	-.350	-.350	-.459	-.435	
		(.018)	(.018)	(.019)	(.019)	(.049)	(.049)	
Enrollment			.017	.041	.062		.079	
			(.009)	(.012)	(.037		(.036)	
Enrollment squared/100					-.010			
					(.016)			
Segment 1 (enrollment 36-45)								-12.6
								(3.80)
Segment 2 (enrollment 76-85)								-2.89
								(2.41)
Root MSE	9.36	8.32	8.30	8.40	8.42	8.79	9.10	10.2
R-squared	.048	.249	.252					
N		2,018			2,018		471	302

Notes: "Adapted from Angrist and Lavy (1999). The table reports estimates of equation (6.2.6) in the text using class averages. Standard errors, reported in parentheses, are corrected for within-school correlation." (The missing parenthesis in "(.037" is in the original.)

(a) Compute m_{sc} for enrollments $e_s = 40, 41, 80, 81, 120, 121$. Why does the rule generate a *fuzzy* rather than a sharp RD?

(b) Write the first and second stages of the 2SLS model. Which variables play the roles of the instrument, the running variable and the treatment?

(c) Explain the sign and the movement of the OLS class-size coefficient across columns (1)–(3).

(d) Using column (4) and the text's round slope of $-.25$, how much would a 7-pupil cut in class size raise math scores, in points and in standard deviations of the class-average score (9.6)? Give the t statistics in columns (4) and (8).

(e) Compare the ratio of the standard errors in columns (8) and (5) with $\sqrt{2018/302}$. What does the comparison say?

Notation.

- e_s: grade enrollment in school s (pupils, integer ≥ 1); m_{sc}: class size predicted by the rule; n_{sc}: actual class size; d_s: percent disadvantaged.
- y_{sc}: class-average math score; ρ: effect of one more pupil per class on the score; π_m: first-stage coefficient on m_{sc}; $\text{int}(x)$: integer part of x.

Step — (a) the rule. Substituting each enrollment into the rule:

$$m(40) = \frac{40}{\text{int}(39/40)+1} = \frac{40}{(0)+1} = 40, \qquad m(41) = \frac{41}{\text{int}(40/40)+1} = \frac{41}{(1)+1} = 20.5, \tag{Q6.1}$$

$$m(80) = \frac{80}{\text{int}(79/40)+1} = \frac{80}{(1)+1} = 40, \qquad m(81) = \frac{81}{\text{int}(80/40)+1} = \frac{81}{(2)+1} = 27, \tag{Q6.2}$$

$$m(120) = \frac{120}{\text{int}(119/40)+1} = \frac{120}{(2)+1} = 40, \quad m(121) = \frac{121}{\text{int}(120/40)+1} = \frac{121}{(3)+1} = 30.25. \tag{Q6.3}$$

Predicted class size drops sharply after each multiple of 40. The design is fuzzy because actual class size n_{sc} does not follow the rule exactly (some schools split grades below 40), so crossing a cutoff changes the *expected* class size, not class size itself with certainty.

Step — (b) 2SLS. The first stage and the second stage are

$$n_{sc} = \pi_0 + \pi_1 d_s + \pi_2 e_s + \cdots + \pi_m\, m_{sc} + \xi_{sc}, \tag{Q6.4}$$

$$y_{sc} = \alpha_0 + \alpha_1 d_s + \beta_1 e_s + \cdots + \rho\, n_{sc} + \eta_{sc}. \tag{Q6.5}$$

The instrument is m_{sc} (the role of t_i), the running variable is e_s (the role of x_i, controlled smoothly), and the treatment is n_{sc} (the role of d_i). Identification comes from the jumps in m_{sc} at 40, 80, 120 conditional on smooth functions of e_s.

Step — (c) OLS. Column (1), $+.322$: larger classes are found in larger, more advantaged schools, and weak pupils are often put in smaller classes, so class size is positively correlated with omitted determinants of scores. Controlling for percent disadvantaged removes most of this (.076), and adding enrollment brings it to .019, insignificant. By the OVB logic of Question 2, the OLS coefficients are biased upwards.

Step — (d) magnitudes. The effect of a change $\Delta n = -7$ is $\rho\,\Delta n$. With column (4) and with the text's slope:

$$\Delta y = (-0.230)(-7) = 1.61, \tag{Q6.6}$$

$$\Delta y = (-0.25)(-7) = 1.75, \tag{Q6.7}$$

$$\frac{\Delta y}{\text{s.d.}} = \frac{(1.75)}{(9.6)} = 0.18229, \tag{Q6.8}$$

about $.18\sigma$, close to the STAR estimate of about $.2\sigma$ in Question 1. The t statistics in columns (4) and (8) are

$$t_4 = \frac{(-.230)}{(.092)} = -2.5, \qquad t_8 = \frac{(-.270)}{(.281)} = -0.9609. \tag{Q6.9}$$

Step — (e) precision. The ratio of standard errors and the square root of the ratio of sample sizes are

$$\frac{\widehat{se}_8}{\widehat{se}_5} = \frac{(.281)}{(.113)} = 2.4867, \tag{Q6.10}$$

$$\sqrt{\frac{2018}{302}} = \sqrt{6.6821} = 2.5850. \tag{Q6.11}$$

Standard errors scale with $1/\sqrt{N}$, so shrinking the sample to the ± 3 window costs precision almost exactly in proportion to $\sqrt{N}$: the estimate stays near $-.25$ but becomes insignificant.

Intuition. The rule is an accident of administration that shifts class size up and down independently of the school's quality once enrollment is controlled for smoothly; 2SLS rescales the score jumps at the cutoffs by the class-size jumps, exactly like a Wald estimator.

Ref: MHE §6.2 (pp. 196–202), fuzzy RD is IV; Question 3 for the Wald logic.

Block 2 — Instrumental variables with the CARD data *(Wooldridge Ch. 15 • Questions 7–8)*

Question 7. Wooldridge, Computer Exercise 15.C3.

Source: Wooldridge (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 15, Computer Exercise C3, printed pp. 492–493, reproduced verbatim.

C3 Use the data in CARD for this exercise.

(i) The equation we estimated in Example 15.4 can be written as

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \cdots + u,$$

where the other explanatory variables are listed in Table 15.1. In order for IV to be consistent, the IV for *educ*, *nearc*4, must be uncorrelated with u. Could *nearc*4 be correlated with things in the error term, such as unobserved ability? Explain.

(ii) For a subsample of the men in the data set, an IQ score is available. Regress *IQ* on *nearc*4 to check whether average IQ scores vary by whether the man grew up near a four-year college. What do you conclude?

(iii) Now, regress *IQ* on *nearc*4, *smsa*66, and the 1966 regional dummy variables *reg*662, ..., *reg*669. Are *IQ* and *nearc*4 related after the geographic dummy variables have been partialled out? Reconcile this with your findings from part (ii).

(iv) From parts (ii) and (iii), what do you conclude about the importance of controlling for *smsa*66 and the 1966 regional dummies in the log(*wage*) equation?

Notation.

- IQ_i: IQ score (available for 2,061 of the 3,010 men); $nearc4_i \in \{0,1\}$: grew up near a four-year college; $smsa66_i$: lived in an SMSA in 1966; $reg66k_i$: 1966 region dummies ($k = 2, \ldots, 9$).
- u_i: error of the wage equation; θ: coefficient on *nearc*4; G_i: the geographic dummies.

Code (data `Aula/class-05-06/materials/card.csv`, CR line endings, "." = missing):

```
import pandas as pd, statsmodels.formula.api as smf
card = pd.read_csv("Aula/class-05-06/materials/card.csv", lineterminator="\r", na_values=".")
iq = card.dropna(subset=["IQ"])                                  # n = 2,061
regs = "+".join(f"reg66{i}" for i in range(2, 10))
a = smf.ols("IQ ~ nearc4", iq).fit()                             # part (ii)
b = smf.ols("IQ ~ nearc4 + smsa66 + " + regs, iq).fit()          # part (iii)
print(b.f_test("smsa66=0," + ",".join(f"reg66{i}=0" for i in range(2, 10))))
```

Key output:

```
(ii)  IQ ~ nearc4                          n = 2061, R2 = 0.0059
              coef      se       t        p
Intercept  100.611   0.627
nearc4       2.596   0.745   3.483   0.0005
(iii) IQ ~ nearc4 + smsa66 + reg662..reg669          R2 = 0.0626
Intercept  104.773
nearc4       0.348   0.814   0.427   0.669
smsa66       1.089   0.809      (reg665 -8.475, reg667 -8.394)
F(9, 2050) for the geographic dummies = 13.80, p < 0.001;  corr(nearc4, smsa66) = 0.412
```

Step — (i) exogeneity. Yes, it could. Families that value education, or parents with more schooling and income, may choose to live near colleges; and colleges are in cities, where schools are better and measured ability higher. Any such factor raises wages directly (it is in u) and is correlated with $nearc4$, which would violate $\text{Cov}(nearc4, u) = 0$.

Step — (ii) unconditional. The simple regression gives

$$\widehat{IQ} = 100.611 + 2.596\, nearc4, \qquad t = \frac{(2.5962)}{(0.7455)} = 3.4825, \tag{Q7.1}$$

so men who grew up near a four-year college have IQs about 2.6 points higher on average, a significant difference ($p = 0.0005$). Taken alone, this suggests $nearc4$ is correlated with ability.

Step — (iii) conditional. With the geographic controls,

$$\widehat{IQ} = 104.773 + 0.348\, nearc4 + 1.089\, smsa66 + G_i'\hat{\psi}, \qquad t = \frac{(0.3479)}{(0.8144)} = 0.4272, \tag{Q7.2}$$

with $p = 0.669$: once geography is partialled out, IQ and $nearc4$ are unrelated. The geographic dummies are jointly very significant ($F = 13.80$). Reconciliation by the OVB formula of Question 2: the short coefficient 2.596 equals the long coefficient 0.348 plus (effect of geography on IQ) $\times$ (relation of geography to $nearc4$). Men near colleges are more often urban ($\text{corr}(nearc4, smsa66) = 0.412$) and live in regions with higher average IQ; that, not proximity, produced part (ii).

Step — (iv) implication. $nearc4$ is a plausible instrument only *conditional on* $smsa66$ and the 1966 regional dummies. They must be in the $\log(wage)$ equation (and hence in the first stage), as in Table 15.1; dropping them would let geography-driven ability differences contaminate the IV estimate.

Intuition. Instrument validity is a conditional statement. A regression of a proxy for the omitted factor (IQ) on the instrument, with and without the controls, is a cheap falsification test of that statement.

Ref: W §15.1–15.2 and Example 15.4 (pp. 462–474); MHE §4.1 (pp. 84–102); site example `example1-iv-card`.

Question 8. Wooldridge, Computer Exercise 15.C5.

Source: Wooldridge (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 15, Computer Exercise C5, printed p. 493, reproduced verbatim.

C5 Use the data in CARD for this exercise.

(i) In Table 15.1, the difference between the IV and OLS estimates of the return to education is economically important. Obtain the reduced form residuals, $\hat{v}_2$, from the reduced form regression $educ$ on $nearc4$, $exper$, $exper^2$, $black$, $smsa$, $south$, $smsa66$, $reg662$, …, $reg669$—see Table 15.1. Use these to test whether $educ$ is exogenous; that is, determine if the difference between OLS and IV is statistically significant.

(ii) Estimate the equation by 2SLS, adding $nearc2$ as an instrument. Does the coefficient on $educ$ change much?

(iii) Test the single overidentifying restriction from part (ii).

Notation.

- $\log(wage)_i$ (*lwage*): log hourly wage; $educ_i$: years of schooling; x_i: the exogenous controls of Table 15.1 (*exper*, $exper^2$, *black*, *smsa*, *south*, *smsa*66, *reg*662–*reg*669).
- $nearc4_i$, $nearc2_i$: grew up near a four-year / two-year college (instruments); v_{2i}: reduced-form error; θ: coefficient of u_{1i} on v_{2i}.
- $\hat{u}_i$: 2SLS residuals; R^2_u: R^2 of the overidentification regression; $n = 3010$; q: number of overidentifying restrictions.

Code:

```
from linearmodels.iv import IV2SLS
X  = "exper+expersq+black+smsa+south+smsa66+" + regs
rf = smf.ols("educ ~ nearc4 + " + X, card).fit()                 # reduced form
card["v2hat"] = rf.resid
cf = smf.ols("lwage ~ educ + " + X + " + v2hat", card).fit()      # part (i)
iv2 = IV2SLS.from_formula("lwage ~ 1 + " + X + " + [educ ~ nearc4 + nearc2]",
                          card).fit(cov_type="unadjusted", debiased=True)   # part (ii)
card["u2sls"] = iv2.resids
ov = smf.ols("u2sls ~ nearc4 + nearc2 + " + X, card).fit()        # part (iii)
print(ov.nobs * ov.rsquared, iv2.sargan)
```

Key output:

```
Table 15.1 replicated: OLS educ 0.0747 (0.0035);  IV (nearc4) educ 0.1315 (0.0550)
(i)   reduced form: nearc4 0.320 (0.088), t = 3.64;  R2 = 0.477
      lwage ~ educ + X + v2hat:  educ 0.1315;  v2hat -0.0571 (0.0528), t = -1.081, p = 0.280
      heteroskedasticity-robust (HC1) t on v2hat = -1.101
(ii)  2SLS with nearc4, nearc2:  educ 0.1571 (0.0526)
      first stage: nearc4 0.321 (0.088), nearc2 0.123 (0.077)
(iii) R2 of u2sls on all exogenous variables = 0.00041467;  n*R2 = 1.248;  p = 0.264
      linearmodels Sargan statistic = 1.248, chi2(1)
```

Step — (i) control-function test. Write the structural equation, the reduced form and the projection of u_1 on v_2:

$$lwage_i = \beta_0 + \beta_1 educ_i + x_i'\beta + u_{1i}, \tag{Q8.1}$$

$$educ_i = \pi_0 + \pi_1 nearc4_i + x_i'\pi + v_{2i}, \tag{Q8.2}$$

$$u_{1i} = \theta\, v_{2i} + e_{1i}. \tag{Q8.3}$$

Since *nearc*4 and x_i are exogenous, *educ* is correlated with u_1 only through v_2, so *educ* is exogenous if and only if $\theta = 0$. Substituting (Q8.3) into (Q8.1) and replacing v_2 by $\hat{v}_2$:

$$lwage_i = \beta_0 + \beta_1 educ_i + x_i'\beta + \theta\,\hat{v}_{2i} + \text{error}_i. \tag{Q8.4}$$

The t statistic on $\hat{v}_2$ in (Q8.4) is

$$t = \frac{\hat{\theta}}{\widehat{se}(\hat{\theta})} = \frac{(-0.05706)}{(0.05281)} = -1.0805, \tag{Q8.5}$$

with $p = 0.280$ (robust $t = -1.101$). We do not reject $H_0 : \theta = 0$: although the IV estimate (0.1315) is almost twice OLS (0.0747), the difference is not statistically significant, because the IV estimate is imprecise. (The coefficient on *educ* in (Q8.4) equals the IV estimate, 0.1315, an algebraic identity.)

Step — (ii) adding nearc2. 2SLS with both instruments gives $\hat{\beta}_1 = 0.1571$ (0.0526). The change is

$$0.1571 - 0.1315 = 0.0256, \tag{Q8.6}$$

about half a standard error: a moderate increase, with a slightly smaller standard error. *nearc*2 has a smaller and less precise first-stage coefficient (0.123, se 0.077) than *nearc*4 (0.321, se 0.088), so it adds little.

Step — (iii) overidentification. Regress the 2SLS residuals on all exogenous variables (both instruments and x_i); under H_0 that all instruments are exogenous, $nR_u^2 \sim \chi_q^2$ with $q = 2 - 1 = 1$ (instruments minus endogenous regressors). Substituting:

$$nR_u^2 = (3010)(0.00041467) = 1.24816, \tag{Q8.7}$$

below the 5% critical value $\chi^2_{1,0.95} = 3.841$ ($p = 0.264$). We do not reject the overidentifying restriction: the two instruments give estimates (0.1315 alone, 0.1571 together) that are statistically compatible. The test only checks that the instruments agree; it cannot detect a bias they share.

Intuition. The Hausman test asks whether OLS and IV differ by more than sampling noise; with a weak-ish instrument the IV noise is large, so even a big point difference is not significant. The overidentification test asks whether two instruments tell the same story.

Ref: W §15.5 (pp. 481–484), testing for endogeneity and overidentifying restrictions; Table 15.1 (p. 474); MHE §4.1 (pp. 84–102).

Block 3 — Difference-in-differences *(Wooldridge Ch. 13 • Questions 9–11)*

Question 9. Wooldridge, Problem 13.2.

Source: Wooldridge (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 13, Problem 2, printed p. 425, reproduced verbatim. Equation (13.9), which the problem refers to, is reproduced from p. 410 for reference.

2 Using the data in KIELMC, the following equations were estimated using the years 1978 and 1981:

$$\widehat{\log(price)} = \underset{(.26)}{11.49} - \underset{(.058)}{.547}\, nearinc + \underset{(.080)}{.394}\, y81 \cdot nearinc$$
$$n = 321,\ R^2 = .220$$

and

$$\widehat{\log(price)} = \underset{(.27)}{11.18} + \underset{(.044)}{.563}\, y81 - \underset{(.067)}{.403}\, y81 \cdot nearinc$$
$$n = 321,\ R^2 = .337.$$

Compare the estimates on the interaction term $y81 \cdot nearinc$ with those from equation (13.9). Why are the estimates so different?

For reference, equation (13.9), p. 410:

$$\widehat{\log(price)} = \underset{(.31)}{11.29} + \underset{(.045)}{.457}\, y81 - \underset{(.055)}{.340}\, nearinc - \underset{(.083)}{.063}\, y81 \cdot nearinc, \qquad n = 321,\ R^2 = .409. \tag{13.9}$$

Notation.

- $\log(price)_i$: log selling price of house i; $y81_i$: sold in 1981 (after the incinerator was announced); $nearinc_i$: within three miles of the incinerator site.
- μ_{gt}: mean log price in cell $g \in \{\text{far}, \text{near}\}$, $t \in \{78, 81\}$; $\beta_0, \delta_0, \beta_1, \delta_1$: the coefficients of (13.9) on the constant, $y81$, $nearinc$, $y81 \cdot nearinc$.
- n_{gt}: number of sales in cell (g, t); in KIELMC $n_{\text{far},78} = 123$, $n_{\text{far},81} = 102$, $n_{\text{near},78} = 56$, $n_{\text{near},81} = 40$.

Step — cell means from (13.9). (13.9) has four parameters for four cells, so it fits the cell means exactly:

$$\mu_{\text{far},78} = \beta_0, \quad \mu_{\text{far},81} = \beta_0 + \delta_0, \quad \mu_{\text{near},78} = \beta_0 + \beta_1, \quad \mu_{\text{near},81} = \beta_0 + \delta_0 + \beta_1 + \delta_1. \tag{Q9.1}$$

Step — first equation (no $y81$). Its regressors define three groups: far houses in both years (constant only), near houses in 1978 (constant + $nearinc$) and near houses in 1981 (all three). OLS on group dummies fits group means, so the interaction coefficient is

$$\hat{\theta}_1 = \mu_{\text{near},81} - \mu_{\text{near},78}. \tag{Q9.2}$$

Substituting (Q9.1) into (Q9.2):

$$\hat{\theta}_1 = (\beta_0 + \delta_0 + \beta_1 + \delta_1) - (\beta_0 + \beta_1) \tag{Q9.3}$$
$$= \delta_0 + \delta_1 = (.457) + (-.063) = .394, \tag{Q9.4}$$

exactly the reported .394. The intercept is the pooled far mean, $11.29 + .457\,(102/225) = 11.4972 \approx 11.49$.

Step — second equation (no *nearinc*). Now the groups are all 1978 sales, far 1981 sales and near 1981 sales, so

$$\hat{\theta}_2 = \mu_{\text{near},81} - \mu_{\text{far},81}. \tag{Q9.5}$$

Substituting (Q9.1) into (Q9.5):

$$\hat{\theta}_2 = (\beta_0 + \delta_0 + \beta_1 + \delta_1) - (\beta_0 + \delta_0) \tag{Q9.6}$$
$$= \beta_1 + \delta_1 = (-.340) + (-.063) = -.403, \tag{Q9.7}$$

again exactly the reported value. The intercept is the pooled 1978 mean, $11.29 - .340\,(56/179) = 11.1836 \approx 11.18$.

Step — why they differ. Each short equation imposes a false restriction. The first sets $\delta_0 = 0$ (no general change in prices from 1978 to 1981), so its interaction is the change in prices of near houses, which mixes the aggregate appreciation (+.457) with the incinerator effect. The second sets $\beta_1 = 0$ (near and far houses equally priced before the incinerator), so its interaction is the 1981 near–far gap, which mixes the pre-existing location discount (−.340) with the effect. Only (13.9), with both controls, isolates the difference-in-differences $\hat{\delta}_1 = -.063$. (Fitting the three equations to KIELMC reproduces all coefficients; the intercept standard errors are .026 and .027 in the data, so the printed (.26) and (.27) appear to have lost a zero.)

Intuition. A DiD regression needs both a group dummy and a period dummy; dropping either one lets the interaction soak up the omitted level difference, a textbook case of omitted-variable bias with dummies.

Ref: W Example 13.3 (pp. 407–410) and Table 13.3 (p. 411); MHE §5.2 (pp. 169–171).

Question 10. Wooldridge, Problem 13.6.

Source: Wooldridge (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 13, Problem 6, printed p. 426, reproduced verbatim.

6 In 1985, neither Florida nor Georgia had laws banning open alcohol containers in vehicle passenger compartments. By 1990, Florida had passed such a law, but Georgia had not.

(i) Suppose you can collect random samples of the driving-age population in both states, for 1985 and 1990. Let *arrest* be a binary variable equal to unity if a person was arrested for drunk driving during the year. Without controlling for any other factors, write down a linear probability model that allows you to test whether the open container law reduced the probability of being arrested for drunk driving. Which coefficient in your model measures the effect of the law?

(ii) Why might you want to control for other factors in the model? What might some of these factors be?

(iii) Now, suppose that you can only collect data for 1985 and 1990 at the county level for the two states. The dependent variable would be the fraction of licensed drivers arrested for drunk driving during the year. How does this data structure differ from the individual-level data described in part (i)? What econometric method would you use?

Notation.

- $arrest_i \in \{0, 1\}$: person i arrested for drunk driving in the year; $y90_i$: observed in 1990; FL_i: lives in Florida (treated state); Georgia is the control.
- δ_1: effect of the open-container law on the arrest probability; x: other factors; $arrrate_{ct}$: fraction of licensed drivers arrested in county c, year t; a_c: county effect.

Step — (i) the model. The linear probability model is

$$arrest_i = \beta_0 + \delta_0\, y90_i + \beta_1\, FL_i + \delta_1\, y90_i \cdot FL_i + u_i. \tag{Q10.1}$$

Taking expectations of (Q10.1) in the four state–year cells and differencing as in Question 4:

$$\delta_1 = \big[P(FL,90) - P(FL,85)\big] - \big[P(GA,90) - P(GA,85)\big], \tag{Q10.2}$$

where $P(s,t)$ is the arrest probability in state s, year t. δ_1 measures the effect of the law; test $H_0 : \delta_1 = 0$ against $H_1 : \delta_1 < 0$ with the t statistic on the interaction (heteroskedasticity-robust, since an LPM is heteroskedastic).

Step — (ii) controls. The samples are independent cross sections, so their composition can change differently in the two states between 1985 and 1990; then (Q10.2) would mix the law's effect with composition effects. Candidates: age, sex, income, education, urban residence, and state-level factors that may have changed differently, such as police enforcement and sobriety checkpoints, beer taxes, the legal blood-alcohol limit or the drinking age. Controls that do not change differently still help by lowering the residual variance and the standard error of $\hat{\delta}_1$.

Step — (iii) county data. Now the unit is the county, observed in both years: a two-period panel, and the outcome is a proportion. Write

$$arrrate_{ct} = \beta_0 + \delta_0\, y90_t + \delta_1\, y90_t \cdot FL_c + x_{ct}'\gamma + a_c + u_{ct}, \tag{Q10.3}$$

where a_c is a time-constant county effect. First-differencing (Q10.3) between 1990 and 1985:

$$\Delta arrrate_c = \delta_0 + \delta_1\, FL_c + \Delta x_c'\gamma + \Delta u_c, \tag{Q10.4}$$

which removes a_c and is estimated by OLS on one observation per county (first-differencing, equal to fixed effects with two periods). Because the outcome is a fraction, weight by the number of licensed drivers, and since the law varies only at the state level with just two states, standard errors should allow for state-level correlation and be read with caution.

Intuition. The individual design compares the change in Florida with the change in Georgia across different people; the county design follows the same counties, so differencing removes their permanent differences.

Ref: W §13.2 (pp. 406–412), policy analysis with pooled cross sections; §13.3–13.4, two-period panels and first differencing.

Question 11. Wooldridge, Computer Exercise 13.C3.

Source: Wooldridge (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 13, Computer Exercise C3, printed p. 427, reproduced verbatim.

C3 Use the data in KIELMC for this exercise.

(i) The variable *dist* is the distance from each home to the incinerator site, in feet. Consider the model

$$\log(price) = \beta_0 + \delta_0 y81 + \beta_1 \log(dist) + \delta_1 y81 \cdot \log(dist) + u.$$

If building the incinerator reduces the value of homes closer to the site, what is the sign of δ_1? What does it mean if $\beta_1 > 0$?

(ii) Estimate the model from part (i) and report the results in the usual form. Interpret the coefficient on $y81 \cdot \log(dist)$. What do you conclude?

(iii) Add age, age^2, $rooms$, $baths$, $\log(intst)$, $\log(land)$, and $\log(area)$ to the equation. Now, what do you conclude about the effect of the incinerator on housing values?

(iv) Why is the coefficient on $\log(dist)$ positive and statistically significant in part (ii) but not in part (iii)? What does this say about the controls used in part (iii)?

Notation.

- $\log(price)_i$: log selling price; $\log(dist)_i$: log distance to the incinerator site (feet); $y81_i$: sold in 1981.
- β_1: price–distance elasticity in 1978; $\beta_1 + \delta_1$: the elasticity in 1981; *intst*: distance to the interstate; *land*, *area*: lot and house area (square feet).

Code (KIELMC from the `wooldridge` package, saved as `Listas/data/kielmc.csv`):

```
k  = pd.read_csv("Listas/data/kielmc.csv")
m2 = smf.ols("lprice ~ y81 + ldist + y81ldist", k).fit()                          # part (ii)
m3 = smf.ols("lprice ~ y81 + ldist + y81ldist + age + agesq + rooms + baths"
             " + lintst + lland + larea", k).fit()                                # part (iii)
```

Key output:

```
(ii)  lprice ~ y81 + ldist + y81ldist          n = 321, R2 = 0.396
             coef      se       t        p
Intercept   8.058    0.508
y81        -0.011    0.805
ldist       0.317    0.052    6.15
y81ldist    0.048    0.082    0.589    0.556
(iii) + age, agesq, rooms, baths, lintst, lland, larea          R2 = 0.787
y81        -0.2254   0.4947
ldist       0.0009   0.0446   0.021
y81ldist    0.0625   0.0503   1.242    0.215
lintst     -0.0600;  larea 0.3507;  corr(ldist, lintst) = 0.781
```

Step — (i) signs. From the model, the price–distance elasticity is

$$\frac{\partial \log(price)}{\partial \log(dist)} = \beta_1 + \delta_1\, y81, \tag{Q11.1}$$

equal to β_1 in 1978 and $\beta_1 + \delta_1$ in 1981. If the incinerator lowers the value of nearby homes, being farther away becomes worth more after it is built, so the elasticity rises: $\delta_1 > 0$. $\beta_1 > 0$ means that already in 1978, before the incinerator was rumored, homes farther from the future site sold for more: the site was in a less valuable area (as Example 13.3 found with *nearinc*).

Step — (ii) estimates. The results in the usual form:

$$\widehat{\log(price)} = \underset{(0.508)}{8.058} - \underset{(0.805)}{0.011}\, y81 + \underset{(0.052)}{0.317}\log(dist) + \underset{(0.082)}{0.048}\, y81 \cdot \log(dist), \quad n = 321,\ R^2 = 0.396. \tag{Q11.2}$$

By (Q11.1), after the incinerator a 1% greater distance adds about 0.048% more to the price than before. The t statistic is $(0.0482)/(0.0818) = 0.589$ ($p = 0.556$): the sign is as predicted but there is no statistical evidence of an incinerator effect.

Step — (iii) with controls. With the house characteristics, $\hat{\delta}_1 = 0.0625$ (0.0503), $t = 1.242$, $p = 0.215$. The estimate is larger but still not significant at conventional levels, so this specification, which uses distance rather than the *nearinc* dummy, gives no firm evidence that the incinerator lowered housing values (contrast the $-.132$, $t \approx -2.53$, that Wooldridge reports with *nearinc* and the same controls).

Step — (iv) the role of the controls. In (ii) $\log(dist)$ has coefficient 0.317 with $t = 6.15$; in (iii) 0.0009 with $t = 0.021$. Homes far from the site differ systematically: they are larger, on larger lots, newer, and farther from the interstate ($\text{corr}(\log dist, \log intst) = 0.781$). In (ii) $\log(dist)$ picks up these characteristics (omitted-variable bias); in (iii) they are controlled for and the pure location effect of distance vanishes. So the controls capture what made distant homes more expensive before the incinerator, which is exactly why they belong in the DiD regression.

Intuition. A continuous treatment intensity (distance) replaces the near/far dummy; the DiD logic is the same, but now the identifying variation is the change in the price–distance gradient.

Ref: W Example 13.3 (pp. 407–410); Kiel and McClain (1995); MHE §5.2 (pp. 169–181), regression DD with covariates.

Block 4 — Bayesian treatment models *(Greenberg Ch. 11 • Questions 12–15)*

Context for Questions 12–15 (Greenberg 2007, §11.1, printed pp. 169–172; the model and the results the exercises refer to, reproduced verbatim, selected passages). The response is related to the covariates and the treatment assignment through

$$y_{0i} = x_i'\beta_0 + u_{0i}, \qquad y_{1i} = x_i'\beta_1 + u_{1i}, \tag{11.1}$$

"where x_i is a K_1 vector of covariates, a 0 subscript indicates assignment to the control group, and 1 indicates assignment to the treatment group." Assignment is modeled by

$$s_i^* = x_i'\gamma_1 + z_i'\gamma_2 + v_i, \qquad s_i = \begin{cases} 1, & \text{if } s_i^* > 0, \\ 0, & \text{if } s_i^* \le 0, \end{cases} \tag{11.2}$$

$$y_i = (1 - s_i)y_{0i} + s_i y_{1i}, \tag{11.3}$$

"where z_i is a $K_2 \times 1$ vector of instrumental variables."

$$\mathrm{Cov}(u_{0i}, v_i) \equiv \Sigma_0 = \begin{pmatrix} \sigma_{00} & \sigma_{0v} \\ \sigma_{0v} & 1 \end{pmatrix}, \qquad \mathrm{Cov}(u_{1i}, v_i) \equiv \Sigma_1 = \begin{pmatrix} \sigma_{11} & \sigma_{1v} \\ \sigma_{1v} & 1 \end{pmatrix},$$

"where $\sigma_{vv} = 1$ because of the probit specification." "Let $\beta = (\beta_0', \beta_1', \gamma_1', \gamma_2')'$, $\gamma = (\gamma_1', \gamma_2')'$, $X_{ji} = \begin{pmatrix} (1-j)x_i' & jx_i' & 0 & 0 \\ 0 & 0 & x_i' & z_i' \end{pmatrix}$, [...] $N_0 = \{i : s_i = 0\}$, $N_1 = \{i : s_i = 1\}$, and n_0 and n_1, respectively, are the number of observations in N_0 and N_1. We assume a Gaussian prior distribution for $\beta \sim N_p(b_0, B_0)$, where $p = 3K_1 + K_2$. [...] we reparameterize the variances in Σ_j as $\omega_{jj} = \sigma_{jj} - \sigma_{jv}^2$ [...]. We define $\psi_j = (\omega_{jj}, \sigma_{jv})$ and specify the prior distributions $\pi(\psi_j) = \pi(\omega_{jj})\pi(\sigma_{jv} \mid \omega_{jj}) = \mathrm{IG}(\nu_{j0}/2, d_{j0}/2)N(m_{j0}, \omega_{jj}M_{j0})$."

$$y_{ji} \mid s_i^*, \beta_j, \gamma, \psi_j \sim N(x_i'\beta_j + \sigma_{jv}e_i, \omega_{jj}), \tag{11.4}$$

"where $e_i = s_i^* - x_i'\gamma_1 - z_i'\gamma_2$. Let Y_j, X_j, Z_j, S_j^*, and E_j, respectively, contain the values of y_{ji}, x_i, z_i, s_i^*, and e_i for $i \in N_j$. Then"

$$Y_j \mid S_j^*, \beta_j, \gamma, \psi_j \sim N_{n_j}(X_j\beta_j + \sigma_{jv}E_j,\ \omega_{jj}I_{n_j}). \tag{11.5}$$

$$E(Y_j \mid S_j^*, \beta_j, \gamma, \omega_{jj}) = X_j\beta_j + m_{j0}E_j, \tag{11.6}$$

$$\mathrm{Var}(Y_j \mid S_j^*, \gamma, \omega_{jj}) = \omega_{jj}(I_{n_j} + E_jM_{j0}E_j'). \tag{11.7}$$

$$f(y_{ji}, s_i = j \mid x_i, z_i, \beta, \psi_j) = N(y_{ji} \mid x_i'\beta_j, \sigma_{jj})\,\Phi\left((2j-1)\frac{\mu_{ji}}{\phi_j}\right). \tag{11.8}$$

Algorithm 11.1: Gibbs algorithm for treatment model (Chib, in press-a)

11.1 Sample ψ_j conditioned on (y, s^*, X, Z, β)

(a) Drawing ω_{jj} marginalized over σ_{jv} from $\mathrm{IG}\left(\frac{\nu_{j0}+n_j}{2}, \frac{d_{j0}+d_j}{2}\right)$, where $d_j = (Y_j - X_j\beta_j - m_{j0}E_j)'(I_{n_j} + E_jM_{j0}E_j')^{-1}(Y_j - X_j\beta_j - m_{j0}E_j)$,

(b) Drawing σ_{jv} conditioned on ω_{jj} from $N(c_j, \omega_{jj}C_j)$, where $c_j = C_j[M_{j0}^{-1}m_{j0} + E_j'(Y_j - X_j\beta_j)]$, $C_j = (M_{j0}^{-1} + E_j'E_j)^{-1}$.

11.2 Sample β conditioned on (y, s^*, ψ_0, ψ_1) from $N_p(\hat\beta, B)$, where $\hat\beta = B(B_0^{-1}b_0 + \sum_{i\in N_0} X_{0i}'\Sigma_0^{-1}y_i^* + \sum_{i\in N_1} X_{1i}'\Sigma_1^{-1}y_i^*)$, $B = (B_0^{-1} + \sum_{i\in N_0} X_{0i}'\Sigma_0^{-1}X_{0i} + \sum_{i\in N_1} X_{1i}'\Sigma_1^{-1}X_{1i})^{-1}$, and $y_i^* = (y_i, s_i^*)'$.

11.3 Sample s^* conditioned on $(y, X, Z, \beta, \psi_0, \psi_1)$ by drawing s_i^* from $\mathrm{TN}_{(-\infty,0)}(\mu_{0i}, \phi_0^2)$ if $s_i = 0$, $\mathrm{TN}_{(0,\infty)}(\mu_{1i}, \phi_1^2)$ if $s_i = 1$, where $\mu_{ji} = x_i'\gamma_1 + z_i'\gamma_2 + \sigma_{jv}\sigma_{jj}^{-1}(y_{ji} - x_i'\beta_j)$, $\phi_j^2 = 1 - \sigma_{jv}^2\sigma_{jj}^{-1}$.

Common notation for the solutions. $j \in \{0, 1\}$: untreated/treated; Φ: standard normal c.d.f.; $\mathrm{IG}(\alpha, \beta)$ has density proportional to $\omega^{-(\alpha+1)}e^{-\beta/\omega}$ $(\omega > 0)$; $\mathrm{TN}_{(a,b)}$: normal truncated to (a, b). The bivariate-normal conditioning rule (Greenberg, Appendix A, eq. A.9) is used throughout: if $(a, b)'$ is bivariate normal,

$$a \mid b \sim N\left(\mu_a + \frac{\sigma_{ab}}{\sigma_{bb}}(b - \mu_b),\ \sigma_{aa} - \frac{\sigma_{ab}^2}{\sigma_{bb}}\right). \tag{A.9}$$

Question 12. Greenberg, Exercise 11.1.

Source: Greenberg (2007), *Introduction to Bayesian Econometrics*, 1st ed., Chapter 11, Exercise 11.1, printed p. 180, reproduced verbatim.

11.1 Verify (11.4).

Notation.

- u_{ji}, v_i: errors of the outcome and assignment equations, jointly $N_2(0, \Sigma_j)$; $\sigma_{jj} > 0$, σ_{jv} real, $\omega_{jj} = \sigma_{jj} - \sigma_{jv}^2 > 0$.
- $e_i = s_i^* - x_i'\gamma_1 - z_i'\gamma_2$: the value of v_i implied by s_i^* and γ.

Step — the ingredients. From (11.1), (11.2) and the covariance matrix Σ_j:

$$y_{ji} = x_i'\beta_j + u_{ji}, \tag{Q12.1}$$

$$s_i^* = x_i'\gamma_1 + z_i'\gamma_2 + v_i, \tag{Q12.2}$$

$$\begin{pmatrix} u_{ji} \\ v_i \end{pmatrix} \sim N_2\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{jj} & \sigma_{jv} \\ \sigma_{jv} & 1 \end{pmatrix} \right), \tag{Q12.3}$$

$$\sigma_{jj} = \omega_{jj} + \sigma_{jv}^2. \tag{Q12.4}$$

Step — conditioning on s^* fixes v. Given s_i^* and γ, solving (Q12.2) for v_i:

$$v_i = s_i^* - x_i'\gamma_1 - z_i'\gamma_2 = e_i. \tag{Q12.5}$$

So conditioning on (s_i^*, γ) is conditioning on $v_i = e_i$.

Step — conditional distribution of u. Apply (A.9) with $a = u_{ji}$, $b = v_i$, and from (Q12.3) $\mu_a = \mu_b = 0$, $\sigma_{aa} = \sigma_{jj}$, $\sigma_{ab} = \sigma_{jv}$, $\sigma_{bb} = 1$. The mean is

$$E(u_{ji} \mid v_i) = (0) + \frac{(\sigma_{jv})}{(1)}(v_i - (0)) \tag{Q12.6}$$

$$= \sigma_{jv} e_i, \tag{Q12.7}$$

using (Q12.5). The variance is

$$\mathrm{Var}(u_{ji} \mid v_i) = (\sigma_{jj}) - \frac{(\sigma_{jv})^2}{(1)} \tag{Q12.8}$$

$$= \omega_{jj}, \tag{Q12.9}$$

by (Q12.4).

Step — back to y. Given $(s_i^*, \beta_j, \gamma, \psi_j)$, $x_i'\beta_j$ in (Q12.1) is a constant, so y_{ji} is u_{ji} shifted by $x_i'\beta_j$. Substituting (Q12.7) and (Q12.9):

$$y_{ji} \mid s_i^*, \beta_j, \gamma, \psi_j \sim N\big(x_i'\beta_j + (\sigma_{jv} e_i),\ (\omega_{jj})\big), \tag{Q12.10}$$

which is (11.4). The distribution depends on $\psi_j = (\omega_{jj}, \sigma_{jv})$ only, which is why the reparameterization was chosen. Independence across i then stacks (Q12.10) into (11.5).

Intuition. Once the latent assignment index is known, the assignment error v_i is known, and the part of the outcome error correlated with it, $\sigma_{jv} e_i$, becomes a regressor: this is a Bayesian control function, the analogue of adding $\hat{v}_2$ in Question 8.

Ref: Greenberg §11.1 (pp. 168–172) and Appendix A (eq. A.9); compare the control-function test of W §15.5 (Question 8).

Question 13. Greenberg, Exercise 11.2.

Source: Greenberg (2007), *Introduction to Bayesian Econometrics*, 1st ed., Chapter 11, Exercise 11.2, printed p. 180, reproduced verbatim.

11.2 Verify (11.6) and (11.7).

Notation.

- Y_j, X_j, E_j: stacked y_{ji}, x_i', e_i for $i \in N_j$ ($n_j \times 1$, $n_j \times K_1$, $n_j \times 1$).
- m_{j0}, $M_{j0} > 0$: prior mean and scale of σ_{jv} given ω_{jj}; η_j: prior deviation of σ_{jv} from m_{j0}; ε_j: the n_j-vector of conditional errors in (11.5).

Step — model and prior. (11.5) and the prior on σ_{jv} are

$$Y_j = X_j\beta_j + \sigma_{jv}E_j + \varepsilon_j, \qquad \varepsilon_j \sim N_{n_j}(0, \omega_{jj}I_{n_j}), \tag{Q13.1}$$

$$\sigma_{jv} = m_{j0} + \eta_j, \qquad \eta_j \sim N(0, \omega_{jj}M_{j0}), \tag{Q13.2}$$

with η_j independent of ε_j given ω_{jj} (prior information versus sampling noise).

Step — marginalize σ_{jv}. Substituting (Q13.2) into (Q13.1):

$$Y_j = X_j\beta_j + (m_{j0} + \eta_j)E_j + \varepsilon_j. \tag{Q13.3}$$

Separating the fixed and random parts of (Q13.3):

$$Y_j = X_j\beta_j + m_{j0}E_j + (\eta_jE_j + \varepsilon_j). \tag{Q13.4}$$

Given (S_j^*, γ), E_j is fixed, so $\eta_jE_j + \varepsilon_j$ is a linear combination of independent normals and is normal.

Step — the mean. Taking expectations of the bracket in (Q13.4):

$$E(\eta_jE_j + \varepsilon_j) = E_j\,E(\eta_j) + E(\varepsilon_j) = E_j(0) + (0) = 0, \tag{Q13.5}$$

so $E(Y_j \mid S_j^*, \beta_j, \gamma, \omega_{jj}) = X_j\beta_j + m_{j0}E_j$, which is (11.6).

Step — the variance. By independence of η_j and ε_j,

$$\begin{aligned} \operatorname{Var}(\eta_jE_j + \varepsilon_j) &= E_j\operatorname{Var}(\eta_j)E_j' + \operatorname{Var}(\varepsilon_j) && \text{(Q13.6)}\\ &= E_j(\omega_{jj}M_{j0})E_j' + (\omega_{jj}I_{n_j}) && \text{(Q13.7)}\\ &= \omega_{jj}(I_{n_j} + E_jM_{j0}E_j'), && \text{(Q13.8)} \end{aligned}$$

substituting the variances from (Q13.1)–(Q13.2) in (Q13.7) and factoring ω_{jj} in (Q13.8). This is (11.7). (The variance does not involve β_j, which is why the book omits it from the conditioning set in (11.7).)

Intuition. Integrating out σ_{jv} turns its prior uncertainty into extra, correlated noise along the direction E_j: observations with large $|e_i|$ get more variance because the control-function term is uncertain.

Ref: Greenberg §11.1 (pp. 170–171); conjugate normal–inverse-gamma regression, Greenberg Ch. 4.

Question 14. Greenberg, Exercise 11.3.

Source: Greenberg (2007), *Introduction to Bayesian Econometrics*, 1st ed., Chapter 11, Exercise 11.3, printed p. 181, reproduced verbatim.

11.3 Verify (11.8).

Notation.

- $f(y_{ji}, s_i = j \mid \cdot)$: joint density of the observed outcome and probability of the observed assignment for unit i in group j; $N(y \mid m, s^2)$: normal density at y.
- μ_{ji}, ϕ_j^2: conditional mean and variance of s_i^* given y_{ji} (defined in step 11.3); $\Phi(-a) = 1 - \Phi(a)$.

Step — factor the joint. Write the joint as marginal times conditional:

$$f(y_{ji}, s_i = j \mid \cdot) = f(y_{ji} \mid \cdot)\, P(s_i = j \mid y_{ji}, \cdot). \tag{Q14.1}$$

Step — the marginal of y. From (Q12.1) and (Q12.3) in Question 12, $u_{ji} \sim N(0, \sigma_{jj})$, so

$$f(y_{ji} \mid \cdot) = N(y_{ji} \mid x_i'\beta_j, \sigma_{jj}). \tag{Q14.2}$$

Step — s^* given y. (y_{ji}, s_i^*) is bivariate normal with means $(x_i'\beta_j,\ x_i'\gamma_1 + z_i'\gamma_2)$, variances $(\sigma_{jj}, 1)$ and covariance $\mathrm{Cov}(u_{ji}, v_i) = \sigma_{jv}$. Apply (A.9) with $a = s_i^*$, $b = y_{ji}$:

$$E(s_i^* \mid y_{ji}) = (x_i'\gamma_1 + z_i'\gamma_2) + \frac{(\sigma_{jv})}{(\sigma_{jj})}(y_{ji} - x_i'\beta_j) = \mu_{ji}, \tag{Q14.3}$$

$$\mathrm{Var}(s_i^* \mid y_{ji}) = (1) - \frac{(\sigma_{jv})^2}{(\sigma_{jj})} = \phi_j^2, \tag{Q14.4}$$

the definitions in step 11.3 of Algorithm 11.1.

Step — the probabilities. For $j = 1$, standardizing with (Q14.3)–(Q14.4):

$$P(s_i = 1 \mid y_{1i}) = P(s_i^* > 0 \mid y_{1i}) = 1 - \Phi\left(\frac{0 - \mu_{1i}}{\phi_1}\right) \tag{Q14.5}$$

$$= \Phi\left(\frac{\mu_{1i}}{\phi_1}\right), \tag{Q14.6}$$

by $1 - \Phi(-a) = \Phi(a)$. For $j = 0$:

$$P(s_i = 0 \mid y_{0i}) = P(s_i^* \le 0 \mid y_{0i}) = \Phi\left(\frac{0 - \mu_{0i}}{\phi_0}\right) = \Phi\left(-\frac{\mu_{0i}}{\phi_0}\right). \tag{Q14.7}$$

Since $2j - 1 = 1$ for $j = 1$ and $2j - 1 = -1$ for $j = 0$, (Q14.6) and (Q14.7) are both

$$P(s_i = j \mid y_{ji}) = \Phi\left((2j-1)\frac{\mu_{ji}}{\phi_j}\right). \tag{Q14.8}$$

Step — assemble. Substituting (Q14.2) and (Q14.8) into (Q14.1):

$$f(y_{ji}, s_i = j \mid x_i, z_i, \beta, \psi_j) = \big(N(y_{ji} \mid x_i'\beta_j, \sigma_{jj})\big)\Big(\Phi\Big((2j-1)\tfrac{\mu_{ji}}{\phi_j}\Big)\Big), \tag{Q14.9}$$

which is (11.8); for $j = 0$ it is the book's $N(y_{0i} \mid x_i'\beta_0, \sigma_{00})\Phi(-\mu_{0i}/\phi_0)$.

Intuition. The likelihood contribution is "probability of the outcome" times "probit probability of the observed assignment, updated by what the outcome reveals about the assignment error"; with $\sigma_{jv} = 0$ it splits into a regression and an ordinary probit.

Ref: Greenberg §11.1 (p. 172), marginal likelihood by Chib's method; §8.2.2 (binary probit with latent data).

Question 15. Greenberg, Exercise 11.4.

Source: Greenberg (2007), *Introduction to Bayesian Econometrics*, 1st ed., Chapter 11, Exercise 11.4, printed p. 181, reproduced verbatim.

11.4 Verify the conditional posterior distributions used in Algorithm 11.1.

Notation.

- $r_j = Y_j - X_j\beta_j - m_{j0}E_j$: residual from the marginal mean (11.6); $A_j = I_{n_j} + E_jM_{j0}E_j'$; $w_j = Y_j - X_j\beta_j$.
- $\epsilon_i = y_i^* - X_{ji}\beta$: the 2×1 error of unit $i \in N_j$, $\epsilon_i \sim N_2(0, \Sigma_j)$; $\propto$: proportional to, as a function of the parameter being sampled.

Step — (a) ω_{jj} marginalized over σ_{jv}. By Bayes, using the marginal density of Y_j from Question 13 and the IG prior:

$$\pi(\omega_{jj} \mid \cdot) \propto N_{n_j}(Y_j \mid X_j\beta_j + m_{j0}E_j,\ \omega_{jj}A_j)\ \pi(\omega_{jj}). \tag{Q15.1}$$

The normal density is proportional to $|\omega_{jj}A_j|^{-1/2}\exp\{-r_j'(\omega_{jj}A_j)^{-1}r_j/2\}$, and $|\omega_{jj}A_j| = \omega_{jj}^{n_j}|A_j|$, so

$$N_{n_j}(\cdot) \propto \omega_{jj}^{-n_j/2}\exp\left\{-\frac{d_j}{2\omega_{jj}}\right\}, \qquad d_j = r_j'A_j^{-1}r_j. \tag{Q15.2}$$

The prior $\mathrm{IG}(\nu_{j0}/2, d_{j0}/2)$ is

$$\pi(\omega_{jj}) \propto \omega_{jj}^{-(\nu_{j0}/2+1)}\exp\left\{-\frac{d_{j0}}{2\omega_{jj}}\right\}. \tag{Q15.3}$$

Substituting (Q15.2) and (Q15.3) into (Q15.1):

$$\pi(\omega_{jj} \mid \cdot) \propto \left(\omega_{jj}^{-n_j/2}e^{-d_j/(2\omega_{jj})}\right)\left(\omega_{jj}^{-(\nu_{j0}/2+1)}e^{-d_{j0}/(2\omega_{jj})}\right) \tag{Q15.4}$$

$$= \omega_{jj}^{-\left(\frac{\nu_{j0}+n_j}{2}+1\right)}\exp\left\{-\frac{d_{j0}+d_j}{2\,\omega_{jj}}\right\}, \tag{Q15.5}$$

the kernel of $\mathrm{IG}\big((\nu_{j0}+n_j)/2, (d_{j0}+d_j)/2\big)$, with d_j as in step 11.1(a).

Step — (b) σ_{jv} given ω_{jj}. The likelihood (11.5) and the prior give

$$\pi(\sigma_{jv} \mid \omega_{jj}, \cdot) \propto \exp\left\{-\frac{(w_j - \sigma_{jv}E_j)'(w_j - \sigma_{jv}E_j)}{2\omega_{jj}}\right\}\exp\left\{-\frac{(\sigma_{jv} - m_{j0})^2}{2\omega_{jj}M_{j0}}\right\}. \tag{Q15.6}$$

Multiplying the exponent of (Q15.6) by $-2\omega_{jj}$ and expanding:

$$\sigma_{jv}^2E_j'E_j - 2\sigma_{jv}E_j'w_j + w_j'w_j + M_{j0}^{-1}(\sigma_{jv}^2 - 2\sigma_{jv}m_{j0} + m_{j0}^2). \tag{Q15.7}$$

Collecting powers of σ_{jv} in (Q15.7):

$$\sigma_{jv}^2(M_{j0}^{-1} + E_j'E_j) - 2\sigma_{jv}(M_{j0}^{-1}m_{j0} + E_j'w_j) + \text{const.} \tag{Q15.8}$$

With $C_j = (M_{j0}^{-1} + E_j'E_j)^{-1}$ and $c_j = C_j(M_{j0}^{-1}m_{j0} + E_j'w_j)$, (Q15.8) is

$$C_j^{-1}(\sigma_{jv} - c_j)^2 + \text{const}, \tag{Q15.9}$$

so $\pi(\sigma_{jv} \mid \omega_{jj}, \cdot) \propto \exp\{-(\sigma_{jv} - c_j)^2/(2\omega_{jj}C_j)\}$, i.e. $N(c_j, \omega_{jj}C_j)$ with $w_j = Y_j - X_j\beta_j$, as in step 11.1(b).

Step — (c) β. For $i \in N_j$ stack $y_i^* = (y_i, s_i^*)'$. By the definition of X_{ji},

$$X_{ji}\beta = \begin{pmatrix}(1-j)x_i'\beta_0 + jx_i'\beta_1 \\ x_i'\gamma_1 + z_i'\gamma_2\end{pmatrix}, \qquad y_i^* = X_{ji}\beta + \epsilon_i, \quad \epsilon_i \sim N_2(0, \Sigma_j), \tag{Q15.10}$$

a SUR system. Prior times likelihood:

$$\pi(\beta \mid \cdot) \propto \exp\left\{-\tfrac{1}{2}(\beta - b_0)'B_0^{-1}(\beta - b_0)\right\}\prod_{j=0}^{1}\prod_{i\in N_j}\exp\left\{-\tfrac{1}{2}(y_i^* - X_{ji}\beta)'\Sigma_j^{-1}(y_i^* - X_{ji}\beta)\right\}. \tag{Q15.11}$$

Multiplying the exponent of (Q15.11) by -2 and collecting terms in β:

$$\beta'\Big(B_0^{-1} + \sum_j\sum_{i\in N_j}X_{ji}'\Sigma_j^{-1}X_{ji}\Big)\beta - 2\beta'\Big(B_0^{-1}b_0 + \sum_j\sum_{i\in N_j}X_{ji}'\Sigma_j^{-1}y_i^*\Big) + \text{const.} \tag{Q15.12}$$

Writing the first bracket as B^{-1} and the second as $B^{-1}\hat{\beta}$, (Q15.12) becomes

$$(\beta - \hat{\beta})' B^{-1} (\beta - \hat{\beta}) + \text{const}, \tag{Q15.13}$$

so $\beta \mid \cdot \sim N_p(\hat{\beta}, B)$ with B and $\hat{\beta}$ exactly as in step 11.2.

Step — (d) s_i^*. Given everything else, s_i^* enters only unit i's joint density and the indicator that ties it to the observed s_i:

$$\pi(s_i^* \mid \cdot) \propto N_2\big(y_i^* \mid X_{ji}\beta, \Sigma_j\big)\big[1(s_i = 0)1(s_i^* \le 0) + 1(s_i = 1)1(s_i^* > 0)\big]. \tag{Q15.14}$$

Factor the bivariate density as $f(y_{ji})f(s_i^* \mid y_{ji})$; $f(y_{ji})$ does not involve s_i^*, and by (Q14.3)–(Q14.4) of Question 14 $f(s_i^* \mid y_{ji}) = N(s_i^* \mid \mu_{ji}, \phi_j^2)$. Substituting into (Q15.14):

$$\pi(s_i^* \mid \cdot) \propto N\big(s_i^* \mid \mu_{ji}, \phi_j^2\big)\big[1(s_i = 0)1(s_i^* \le 0) + 1(s_i = 1)1(s_i^* > 0)\big], \tag{Q15.15}$$

i.e. $\text{TN}_{(-\infty,0)}(\mu_{0i}, \phi_0^2)$ when $s_i = 0$ and $\text{TN}_{(0,\infty)}(\mu_{1i}, \phi_1^2)$ when $s_i = 1$, as in step 11.3 (whether 0 is included is a set of probability zero).

Intuition. Every block is conjugate once the latent s^* is filled in: given s^* the model is a normal SUR regression with a control-function term, so the Gibbs sampler only cycles through inverse-gamma, normal and truncated-normal draws.

Ref: Greenberg §11.1 (pp. 170–172), Algorithm 11.1; §9.1 (pp. 134–139), SUR; §8.2.2, truncated-normal latent data.